



Biology ATP Analysis By Vasumitra Gajbhiye

▼ State one safety precaution that the students would take when evaporating substances containing ethanol.

- not using a Bunsen / flames / AW ;
- using water-bath / electric heater

▼ Suggest why the outside of the dialysis tubing bag is rinsed?

- to remove/ prevent contamination
- To remove (any names substance) from the outside of the bag

▼ How to use a knife carefully?

cut away from body, towards a flat surface like tile or wooden board

▼ How to keep pH constant in enzyme question?

use buffer

▼ How to describe an inverted V graph?

- increase and decrease

- peak at: {coordinates of the peak point **with units**}

▼ What is the hazard when performing test for reducing sugar, protein, starch.

<i>one hazard from:</i>	<i>one precaution from the same row as the hazard:</i>
(damage to skin from) food testing reagents / chemicals / solutions / liquids	goggles / eye protection / gloves
hot water / flame / heat (for Benedict's test)	stand up during practical work / use of a water-bath / (heat-proof) gloves / use of tongs

▼ The students wanted to obtain a more accurate value for the temperature at which amylase works best. Suggest further investigative work that the students should carry out.

- test at smaller intervals of temperature ;
- select a narrower range of temperatures around/ below and above 35°C (35deg C is the peak for the question)

▼ Suggest why repeating the procedure several times would improve the investigation so that anomalous results can be, identified / excluded ;

▼ Suggest why when estimating population of an organism, we use large number of samples?

to give a representative sample / to avoid bias /

▼ In above question 🖐️, what if we took small sample.

Not a representative sample

▼ Why you should use percentage for comparison rather actual value?
different initial value(e.g. mass, population)

▼ Suggest why the student dried the potato cylinders before measuring their mass.
remove any excess solution **so that it is not included in the mass ;**

▼ Suggest why some named factors were same for all experiment?
so that the results were comparable

▼ Improvement for osmosis in potato stick experiment.

- measure mass of cylinders (instead of length) ;
- more intermediate concentrations of sugar solution
- extend the range of concentrations used ;
- repeat at each concentration / replicate the investigation ;
- soak for longer than 20 minutes
- use warmer sugar solutions (to speed up diffusion) ;
- keep all at the same temperature ;
- dry cylinders before measuring ;

▼ Why it is important to stir?

idea of uniform, distribution / concentration (of yeast cells) / AW ;

▼ An improvement for **any** investigation?

Repeat and average

NOTE → Only write this improvement if they have performed the investigation only once.

▼ Source of error for an experiment?

no repeat

▼ How to maintain temperature of water bath?

1. use thermostatically controlled water-baths ;
2. use a Bunsen burner AND thermometer (method outlined) / use a Bunsen burner and monitor the temperature ;
3. top up with, hot water / cold water / ice, and monitor /use a thermometer ;
4. insulate the container / AW ;

▼ How to decrease humidity around plant?

- dehumidifier/ use a desiccant

▼ How to increase humidity around plant?

bowls of hot water around apparatus / place in plastic bag / AW ;

▼ Test for Reducing Sugar.

- Add benedict reagent
- Heat it in water bath at 85°C
- Colour change from blue to green/yellow/orange/red

▼ Test for Protein.

- Add Biuret reagent
- Colour change from Blue to Purple

▼ Test for Vitamin C.

- Add DCPIP solution
- Solution decolourise, i.e. change colour from blue to colorless

▼ Test for Fat.

- Mix sample with ethanol.
- Add this solution to water.
- Positive test if emulsion is seen, i.e. the solution turn milky-cloudy.

▼ Test for Starch.

- Add Iodine **solution**
- Colour change from yellow-brown to blue-black

▼ Error and improvement for any experiment **involving indicators.**

- Error → Judgement of endpoint is subjective and qualitative
- Improvement → Compare to a standard colour chart or colorimeter

▼ Error and improvement for any experiment **involving drops of a liquid.**

- Error → Volume of drops size is inconsistent or unknown or different
- Improvement → Use syringe/ burette/ graduated pipette

▼ Error and improvement for any experiment **involving shaking or stirring a liquid.**

- Error → different degree of, shaking / mixing / stirring

- Improvement → shake at a fixed speed or use a (mechanical) stirrer
- Error → different time of, mixing or shaking
- Improvement → shake or stir, for a fixed time ;

NOTE: There are 3 words that can be used → shaking, mixing or stirring. use the word that is given in the **question** consistently in writing the error and improvement.

▼ Error and improvement for any experiment **involving hot or cold water.**

Error → temperature of the water not controlled / temperature changes

Improvement → thermostatically controlled water-bath / insulation / Bunsen burner

▼ Error and improvement for any experiment **involving unknown volume of a liquid.**

Error → volume of starch not measured

Improvement → measuring cylinder / burette / pipette

▼ What is anomalous result?

Result that does not follow the general pattern or trend shown by other results.

▼ What should be done with the anomalous result.

Exclude the anomalous result while calculating the average. Exclude it while drawing the line of best fit.

▼ Explain why a control group was used in this investigation

to show that the beetroot causes the effect

- Bar graphs or histograms don't peak, use the word 'most common'.
- Always write reading in table to same number of significant figures.
- In starch experiment always write iodine **solution** not just iodine.
- When suggesting improvement regarding drops of some liquid say: measure the volume of substance instead of drops.
- Hydrogen carbonate is purple in low concentration of CO_2 . it is yellow in high concentration of CO_2 . and red in normal air concentration of CO_2 .
- In every 6 marks question mention these points. Example: write a investigation the effect of temperature on activity of amylase:

- use at least 2 or 3 independent variable (use at least 2 or 3 temperature)
- Write how you will maintain the given independent variable (use thermostatically controlled water bath to maintain temperature)
- Repeat the investigation **at least twice more.**
- **Relevant** safety precaution. (use tongs to handle hot test tubes). other examples can be, wear safety goggles, gloves and lab coat. handle enzymes carefully, while using knife cut away from knife. Use whichever precaution suites your investigation.

BEST OF LUCK BUDDY 😊

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